# Binomial Random Variables

## Counting the number of successes

Download the section 8 .Rmd handout to `STAT240/lecture/08-binomial`.

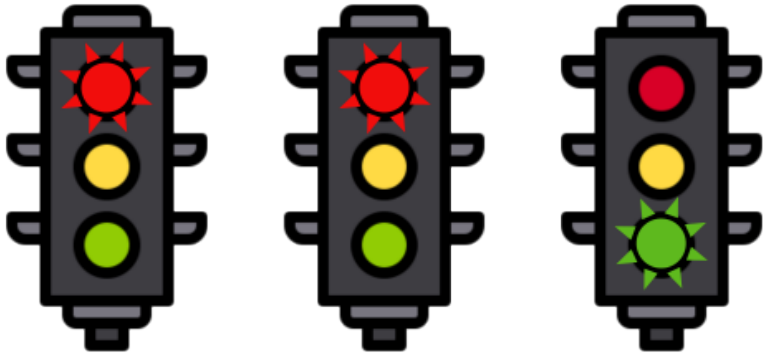
Material in this section is covered by Chapter 10 on the notes website.

How do we decide on a model for a real life population or process?

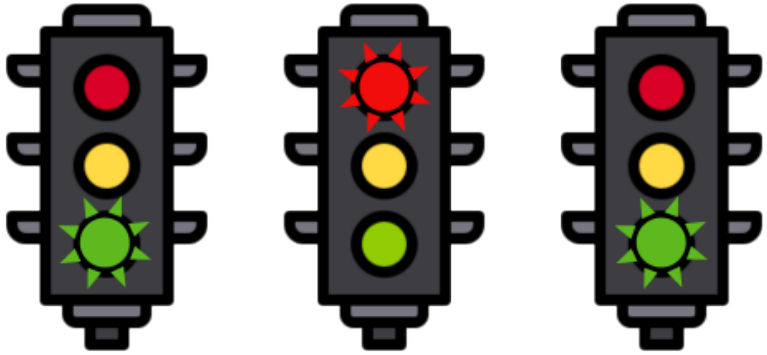
Options:

- Repeat the process infinity times
- Make some **reasonable assumptions**

Let $X$ be the number of green lights I hit out of 3.



x = 1



x = 2

What is the distribution of $X$? Let's assume:

- Lights are independent
- Each has probability 0.6 of being green

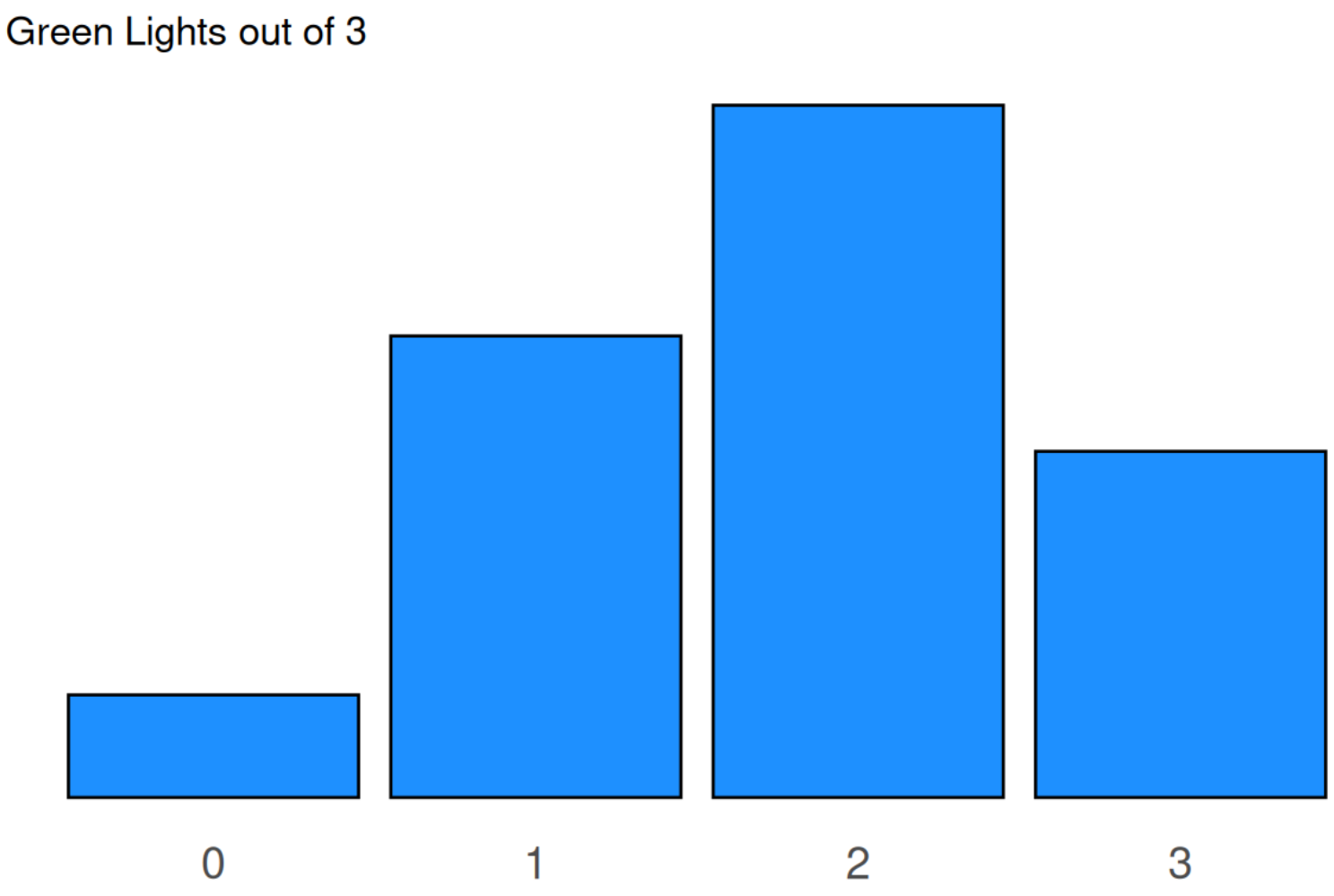
This simplification lets us calculate probabilities.

$X$ = number of green lights.

| Outcomes | $x$ | $P(X = x)$ |
|---|---|---|
| RRR | 0 | $(0.4)(0.4)(0.4) = 0.064$ |
| | 1 | |
| GGR, GRG, RGG | 2 | $3(0.6)(0.6)(0.4) = 0.432$ |
| | 3 | |

Complete the probability distribution of $X$.

| Outcomes | $x$ | $P(X = x)$ |
|---|---|---|
| RRR | 0 | $(0.4)(0.4)(0.4) = 0.064$ |
| | 1 | |
| GGR, GRG, RGG | 2 | $3(0.6)(0.6)(0.4) = 0.432$ |
| | 3 | |


Green Lights out of 3
0
1
2
3

$X$ is a count of "successes" in 3 tries.

$X$ is a **binomial** RV. A binomial counts the number of times a desired outcome occurs in many tries.

We say $X$ has a "binomial distribution".

A binomial is a specific type of discrete RV that has its own rules and formulas.

- Some statistical methods are based on a binomial model

Are these methods useful? We need to assess whether the problem of interest is actually binomial.

The lights (trials) have specific properties:

- **B**: they are binary (success, or failure)
- **I**: they are independent
- **N**: fixed sample size $n$
- **S**: they have the same probability $p$

Do the following situations describe a binomial RV?

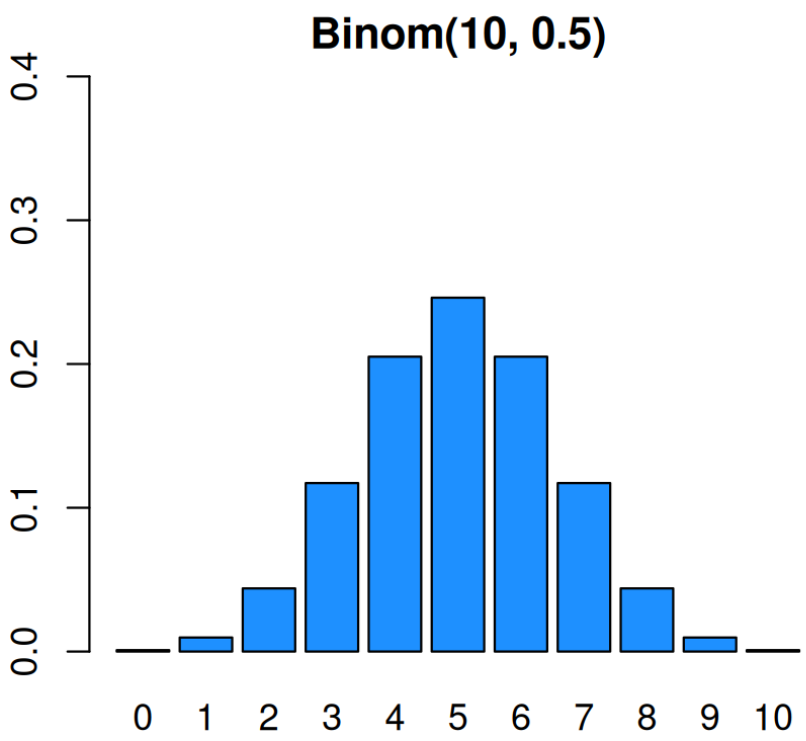
- Let $X$ be the number of green lights we hit before seeing the first red light.
- Let $X$ be the number of students out of 120 who pass a chemistry class.
- Let $X$ be the number of employees out of 30 who show up late to work today.

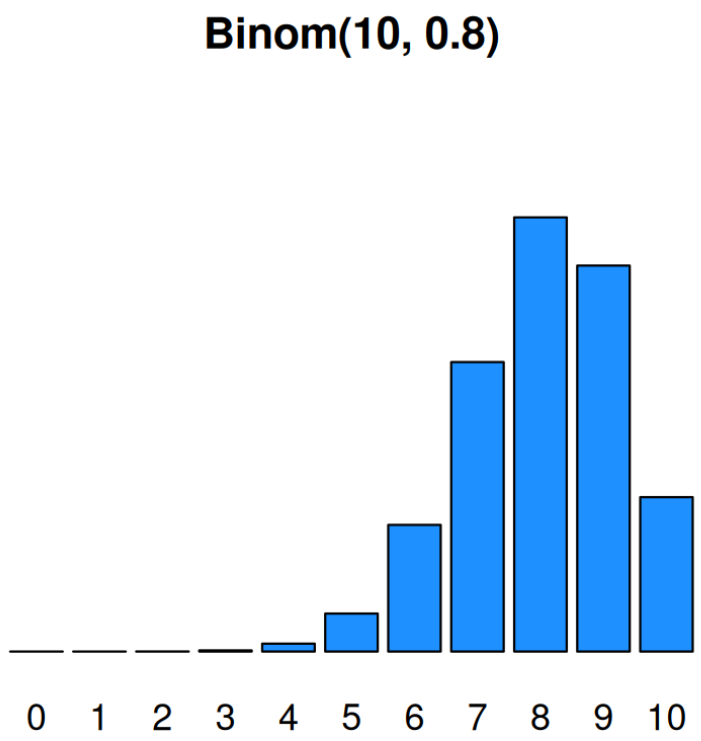
Write $X \sim Binom(n, p)$.

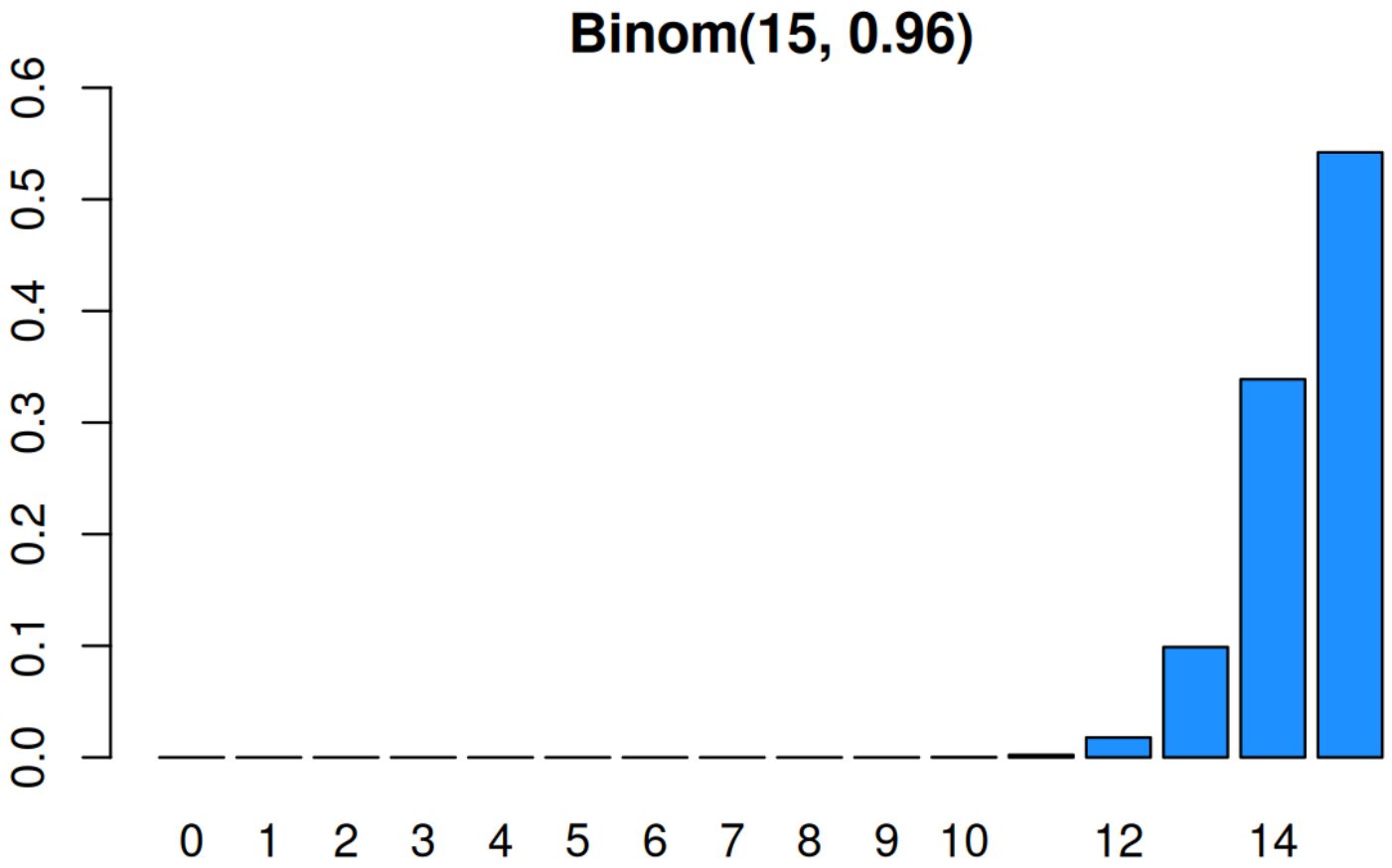
- $n$: pre-determined number of trials
- $p$: individual success probability

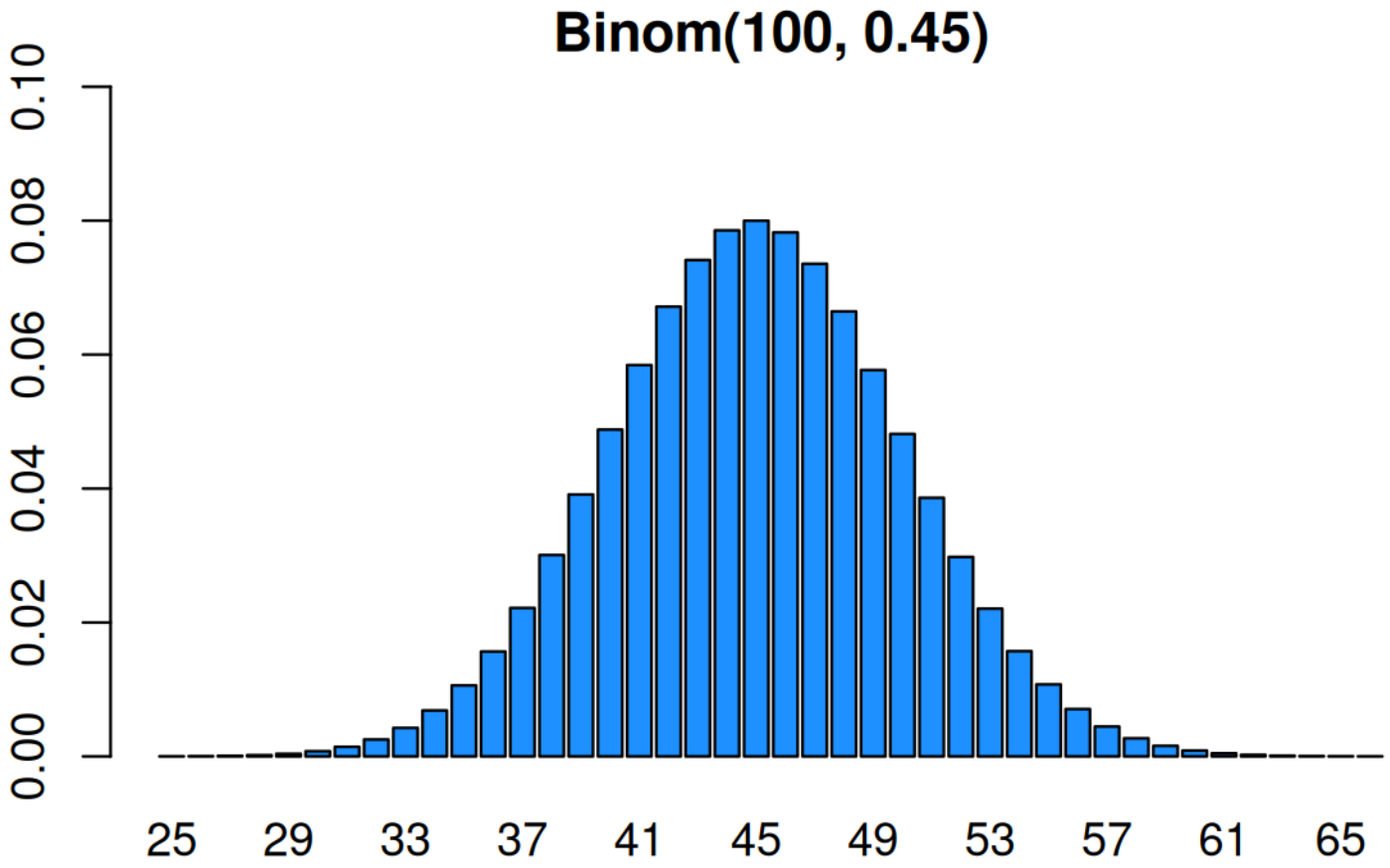
So $X \sim Binom(3, 0.6)$ for the traffic lights.

The shape of the distribution depends on $n$ and $p$.

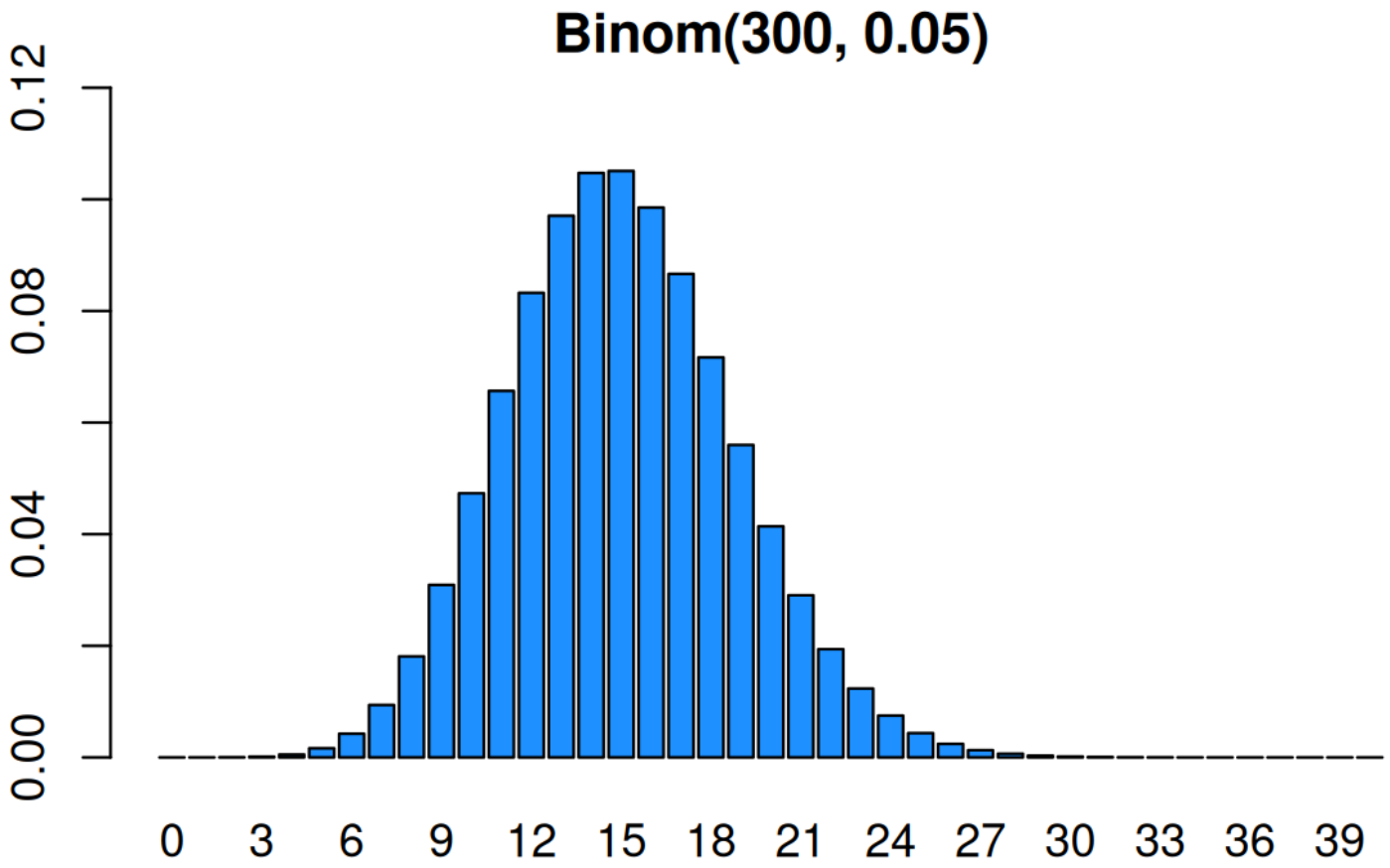

Binom(10, 0.5)
0.0
0.1
0.2
0.3
0.4
0 1 2 3 4 5 6 7 8 9 10


Binom(10, 0.8)
0 1 2 3 4 5 6 7 8 9 10


Binom(15, 0.96)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
1
2
3
4
5
6
7
8
9
10
12
14




(note the zoomed-in x axis)


Binom(300, 0.05)
0.00
0.04
0.08
0.12
0
3
6
9
12
15
18
21
24
27
30
33
36
39

For the traffic lights,

$$P(X = 2) \;=\; 3(0.6)^2(0.4)^1$$

- 2 successes, 1 failure
- 0.6 is the sucess probability
- 0.4 is the failure probability
- 3 is the number of orderings

In general,

$$P(X = x) \;=\; \binom{n}{x} p^x (1-p)^{n-x}$$

- $x$ successes, $n - x$ failures
- $p$ is the sucess probability
- $1 - p$ is the failure probability
- $\binom{n}{x}$ is the number of orderings

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$n!$ is the product of numbers 1 to $n$. $0! = 1$.

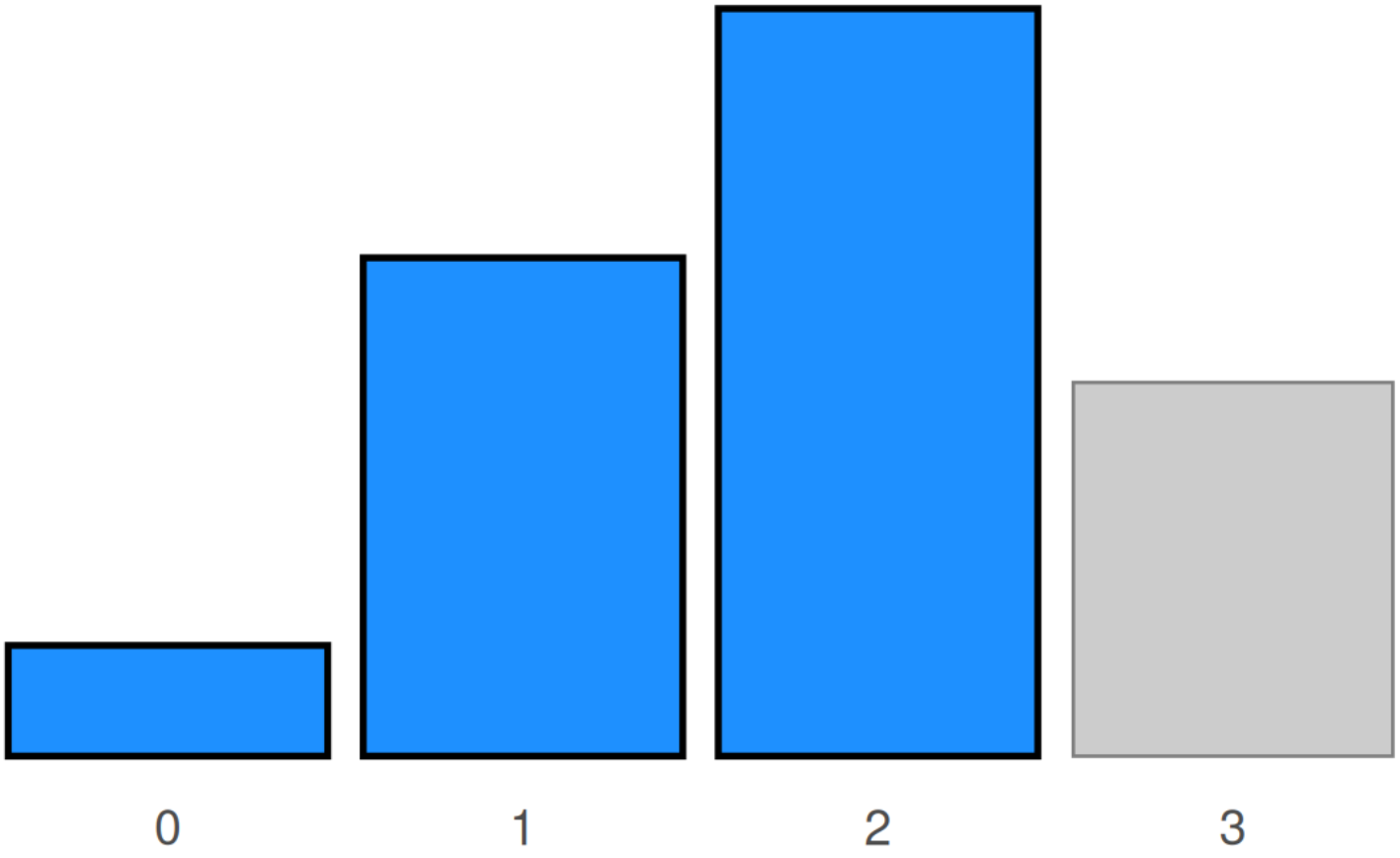
Can be calculated with `choose()` or `factorial()`.

$$P(X = x) \;=\; \binom{n}{x} p^x (1-p)^{n-x}$$

R `dbinom()` calculates this value given $x, n, p$.

`pbinom()` finds a "less than or equal" (cumulative) probability.

$P(X \leq 2)$



`pbinom()` calculates area to the *left*, including the x value specified.

Let $Y \sim Binom(8, 0.3)$. Find the following:

- $P(Y \leq 5)$
- $P(Y > 4)$
- $P(Y \geq 4)$
- $P(3 \leq Y \leq 6)$

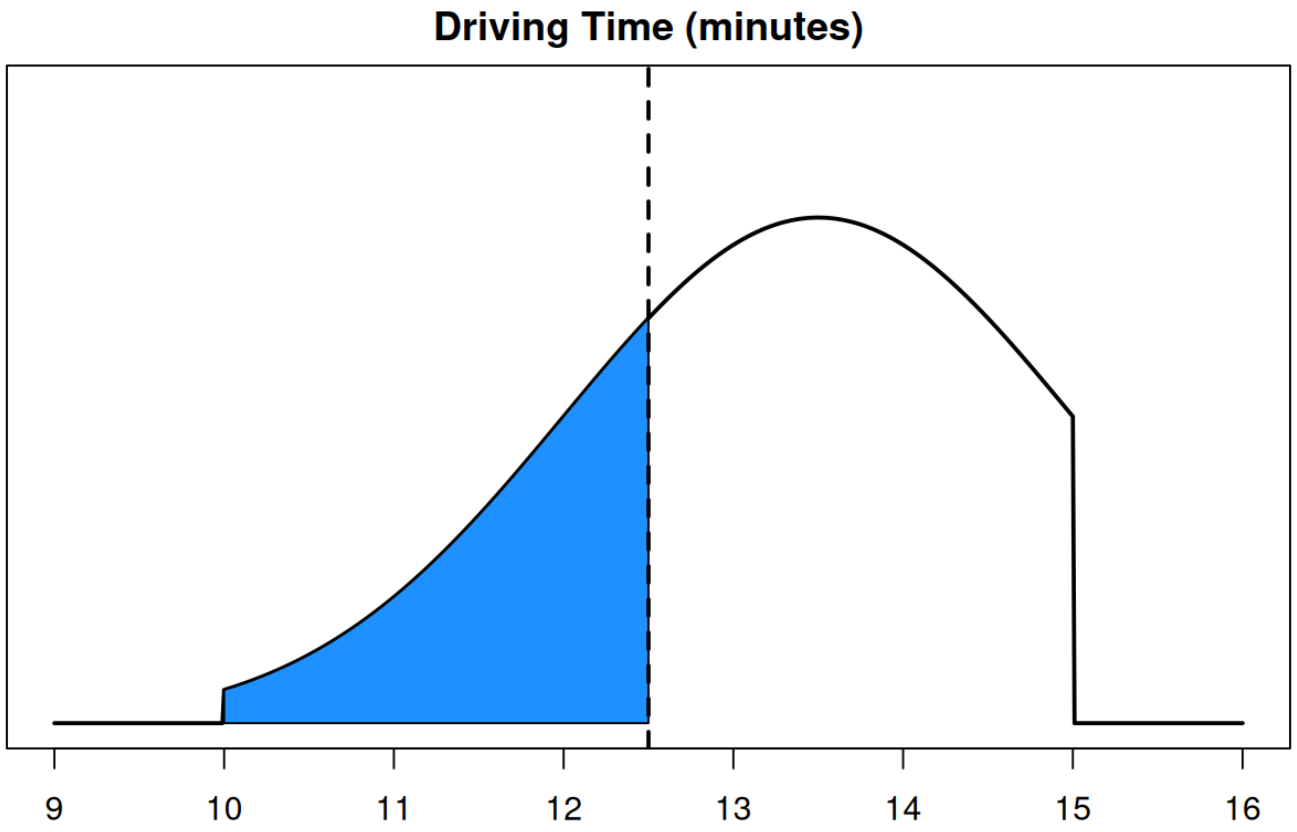
We have shortcuts for the mean/EV and variance of $X \sim Binom(n, p)$.

$$\text{mean } \mu = np, \quad \text{var } \sigma^2 = np(1-p)$$

In the lights example, $\mu = 1.8$ and $\sigma^2 = 0.72$.

Another way to quantify a RV is with a **percentile** or **quantile**.

In a population, 30% of individuals are below the 30th percentile, and 70 are above.

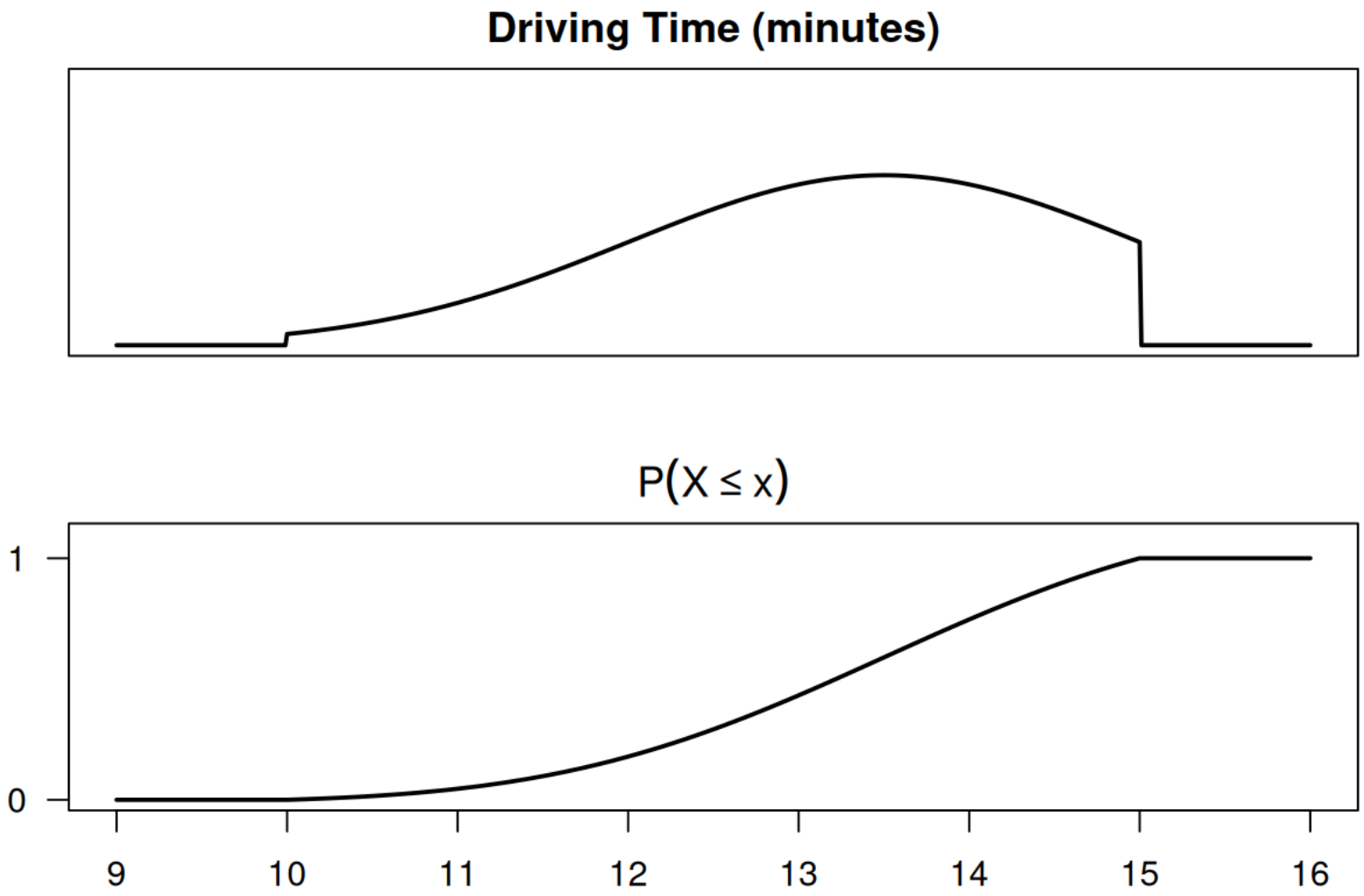


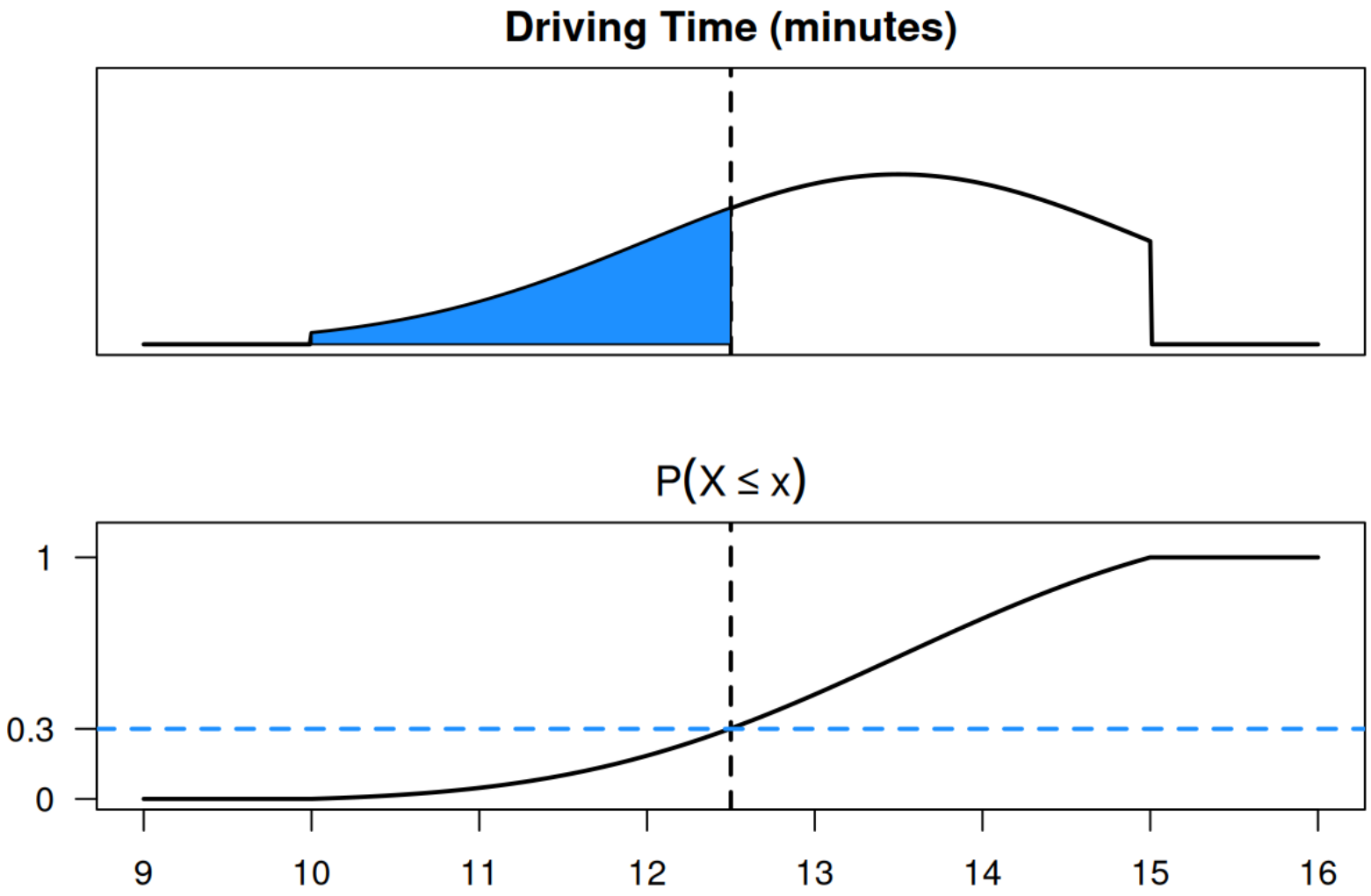

"30% of the time, we get there in less than 12.5 minutes."

In general, quantiles subdivide a specific fraction of the population.

The $p$ percentile is the value such that there is $p$ probability to the left and $1 - p$ to the right.

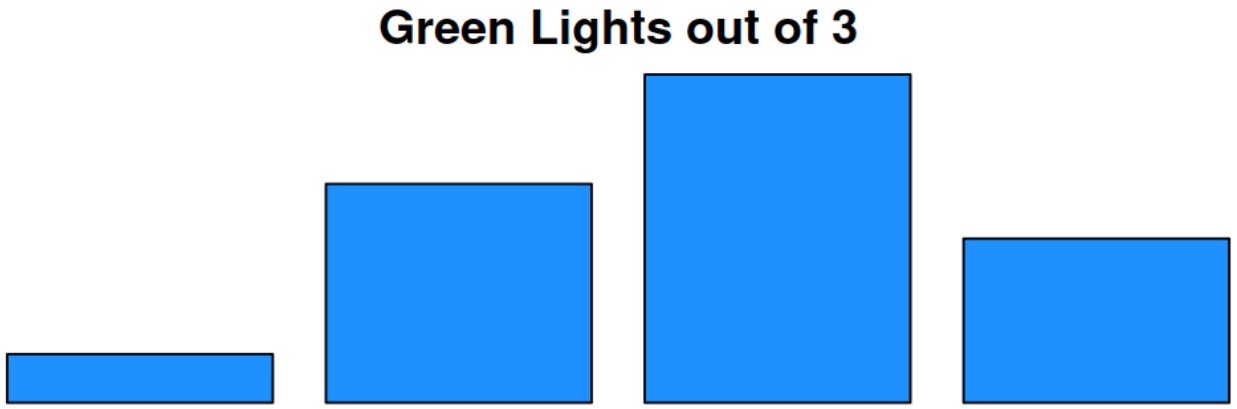
Imagine graphing the *cumulative* probability:

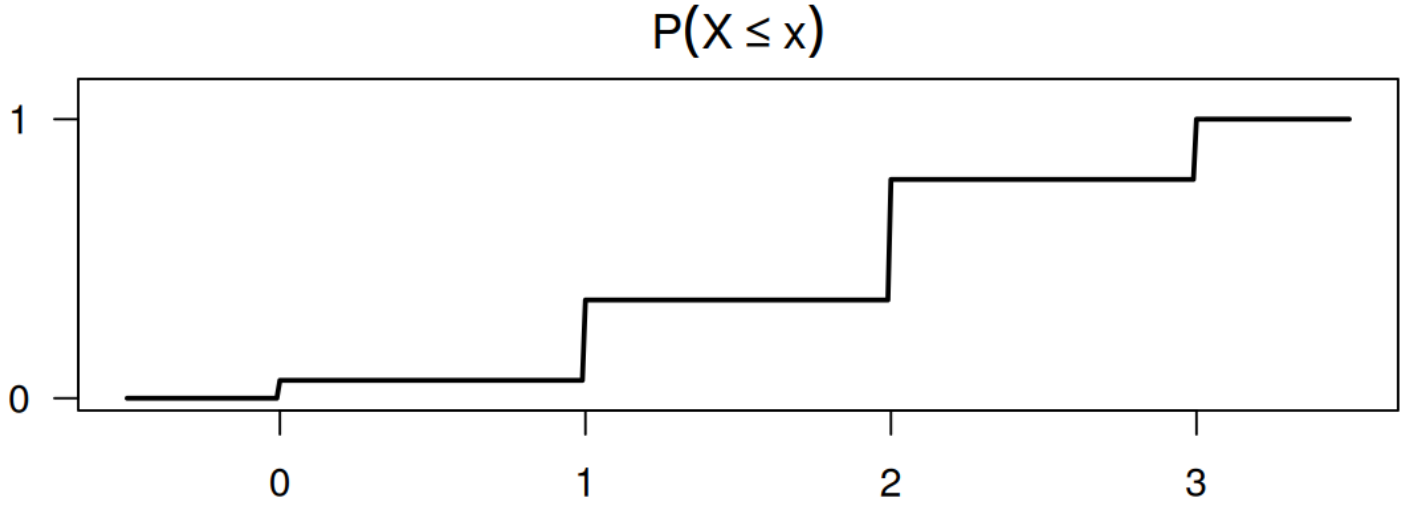


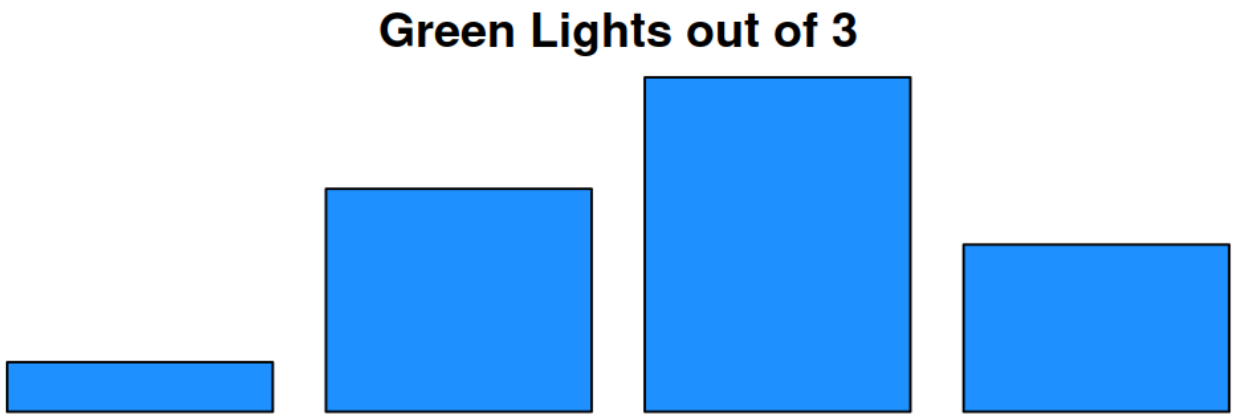


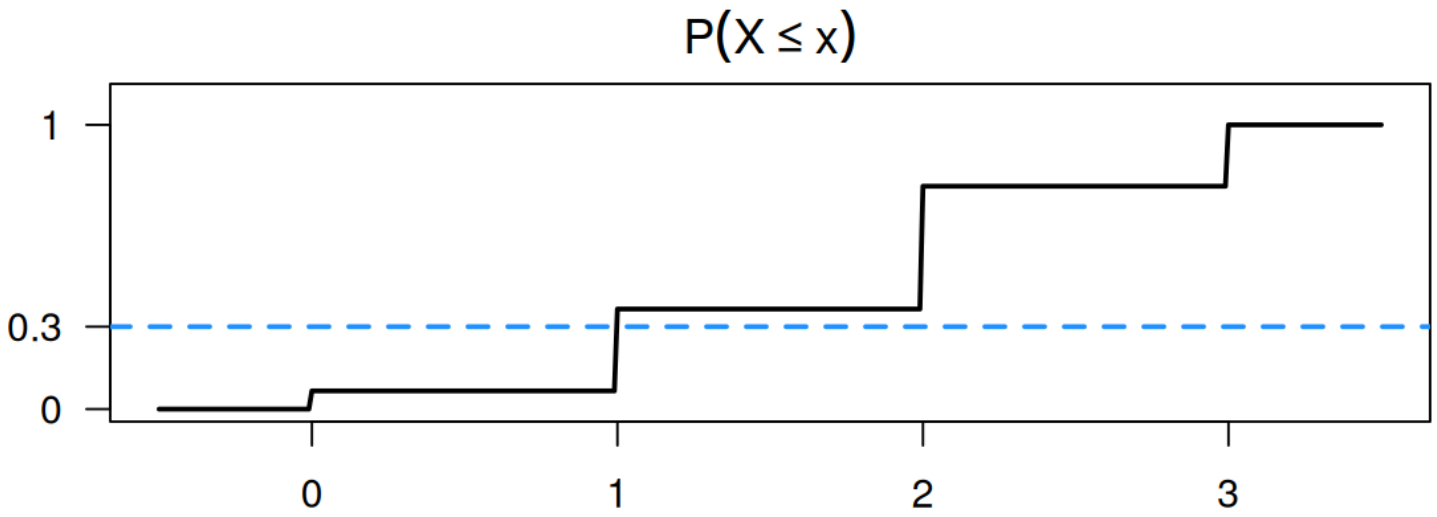

$12.5 = 30\text{th percentile}$

# Percentiles work differently for discrete RVs.






The 30th percentile is 1. Verify with ‘qbinom()‘.

In general, the $p$ percentile of a discrete variable is given by $q$.

How large does $q$ need to be before

$$P(Binom \leq q)$$

is at least $p$?

For binomial RVs, we have `qbinom()` and `pbinom()`.

- `pbinom()`: input $x$ value, output cumulative probability
- `qbinom()`: input cumulative probability, output $x$ value (i.e. $q$)

| **Command** | **In** | **Out** |
|---|---|---|
| `dbinom` | A value $x$ | $P(X = x)$ |
| `pbinom` | A value $x$ | $P(X \leq x)$ |
| `qbinom` | A probability $p$ | $q$ for $P(X \leq q) = p$ |

Let $X \sim Binom(90, 0.7)$.

- Find the mean $\mu$, var $\sigma^2$ and sd $\sigma$ of $X$
- What is $P(X = \mu)$?
- What is $P(\mu - \sigma \leq X \leq \mu + \sigma)$?
- What are the 5th and 95th percentiles of $X$?